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BrightLearn SQL Fundamentals:

Select & Filtering - Exercise 01

## 1. Selecting & Sorting

Table: employees

| id | first name | last name | department | Salary | hire date  | city         |
|----|------------|-----------|------------|--------|------------|--------------|
| 1  | Alice      | Green     | IT         | 70000  | 2000-01-10 | Johannesburg |
| 2  | Brian      | Lee       | HR         | 45000  | 2019-03-22 | Cape Town    |
| 3  | Cathy      | Zulu      | Finance    | 65000  | 2018-07-18 | Durban       |
| 4  | David      | Mokoena   | Marketing  | 50000  | 2001-11-05 | Pretoria     |
| 5  | Eva        | Naidoo    | IT         | 72000  | 2017-09-30 | Johannesburg |

1. Retrieving all columns from employees table.

Select \*

From employees;

2. Find all unique departments.

Select distinct department

From employees;

| Department |
|------------|
| IT         |
| HR         |
| Finance    |
| Marketing  |

3. Retrieve first and last names ordered by salary descending.

| first_name | last_name |
|------------|-----------|
| Eva        | Naidoo    |
| Alice      | Green     |
| Cathy      | Zulu      |
| David      | Motoeng   |
| Brian      | Lee       |

SELECT first\_name, last\_name  
FROM employees  
ORDER BY salary DESC;

4. Retrieve the top 3 highest-paid employees

SELECT id, first\_name, last\_name, salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 3;

| id           | first_name     | last_name         | salary            |
|--------------|----------------|-------------------|-------------------|
| 5            | Eva            | Naidoo            | 72 000            |
| 1            | Alice          | Green             | 70 000            |
| <del>5</del> | <del>Eva</del> | <del>Naidoo</del> | <del>65 000</del> |
| 3            | Cathy          | Zulu              | 65 000            |

Filtering with WHERE

5. Find employees in the IT department.

SELECT id, first\_name, last\_name, department, salary  
FROM employees  
WHERE department = 'IT';

| id | first_name | last_name | department | salary |
|----|------------|-----------|------------|--------|
| 1  | Alice      | Green     | IT         | 70 000 |
| 5  | Eva        | Naidoo    | IT         | 72 000 |



6. Find employees in Finance with salary > 60 000  
 Select id, first\_name, last\_name, department, salary  
 FROM employees  
 WHERE department = 'Finance' AND salary > 60 000;

| id | first_name | last_name | department | salary |
|----|------------|-----------|------------|--------|
| 3  | Cathy      | Zulu      | Finance    | 65 000 |

7. Find employees in HR or Marketing  
 Select id, first\_name, last\_name, department  
 FROM employees  
 WHERE department = 'HR' OR department = 'Marketing';

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 2  | Brian      | Lee       | HR         |
| 4  | David      | Mokoena   | Marketing  |

8. Find employees not in IT  
 Select id, first\_name, last\_name, department  
 FROM employees  
 WHERE NOT department = 'IT';

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 2  | Brian      | Lee       | HR         |
| 3  | Cathy      | Zulu      | Finance    |
| 4  | David      | Mokoena   | Marketing  |

9. Find employees in IT, HR or Finance using IN

SELECT id, first\_name, last\_name, department  
FROM employees

WHERE department IN('IT', 'HR') OR department = 'Finance'

| id | first_name | last_name | department |
|----|------------|-----------|------------|
| 1  | Alice      | Green     | IT         |
| 2  | Brian      | Lee       | HR         |
| 5  | Eva        | Naidoo    | IT         |

10. Find employees in IT with salary > 65 000 and city Johannesburg

SELECT id,

first\_name, last\_name, department, salary, city  
FROM employees

WHERE department = 'IT'

AND salary > 65 000 AND city = 'Johannesburg';

| id | first_name | last_name | department | Salary | City         |
|----|------------|-----------|------------|--------|--------------|
| 1  | Alice      | Green     | IT         | 70 000 | Johannesburg |
| 5  | Eva        | Naidoo    | IT         | 72 000 | Johannesburg |